

CAPSTONE PROJECT REPORT

(Project Term August 2025 – May 2026)

Machine Learning-Based Prediction of Tsunami Occurrence from Seismic Events

Submitted by

Chirag Dhakad	Registration Number: 12221065
Aishwary Gupta	Registration Number: 12221728
Ashimananda Kar	Registration Number: 12216860
Aarav Raj	Registration Number: 12216583

Project Group Number: 2RGC0303

Course Code: CSE439

Under the Guidance of

Dr. Avnish Thakur, Assistant Professor

**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
LOVELY PROFESSIONAL UNIVERSITY
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**TOPIC APPROVAL PERFORMANCE**

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SR.NO.	NAME OF STUDENT	Prov. Regd. No.	BATCH	SECTION
1	Aishwary Gupta	12221728	2022	222TP
2	Chirag Dhakad	12221065	2022	222EG
3	ASHIMANANDA KAR	12216860	2022	222QM
4	Aarav Raj	12216583	2022	322JL

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PAC Committee Members		
PAC Member (HOD/Chairperson) Name: Dr. Harjeet Kaur	UID: 12427	Recommended (Y/N): Yes
PAC Member (Allied) Name: Sudha Shanker Prasad	UID: 21826	Recommended (Y/N): Yes
PAC Member 3 Name: Cherry Khosla	UID: 13436	Recommended (Y/N): Yes

Final Topic Approved by PAC: System to predict Tsunami using Machine Learning models**Overall Remarks:** Approved**PAC CHAIRPERSON Name:** 11265::Dr. Dalwinder Singh**Approval Date:** 04 Dec 2025

DECLARATION

We hereby declare that the project work entitled *Machine Learning-Based Prediction of Tsunami Occurrence from Seismic Events* is an authentic record of our own work carried out as a requirement of the Capstone Project for the award of the B.Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara, under the guidance of Dr. Avnish Thakur, during August 2025 to May 2026. All the information furnished in this Capstone Project Report is based on our own intensive research and study, and is genuine to the best of our knowledge.

Project Group Number: 2RGC0303

Name of Student: **Chirag Dhakad**

Registration No.: 12221065

(Signature)

Date: _____

Name of Student: **Aishwary Gupta**

Registration No.: 12221728

(Signature)

Date: _____

Name of Student: **Ashimananda Kar**

Registration No.: 12216860

(Signature)

Date: _____

Name of Student: **Aarav Raj**

Registration No.: 12216583

(Signature)

Date: _____

CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have successfully completed this Capstone Project under my guidance and supervision. The present work is the result of their original investigation, effort and academic study. No part of the work contained in this report has been submitted for any other degree at any University. This Capstone Project Report is fit for submission and is in partial fulfilment of the conditions for the award of the B.Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara.

Dr. Avnish Thakur

Assistant Professor

School of Computer Science and Engineering,

Lovely Professional University,

Phagwara, Punjab.

Date: _____

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Chirag Dhakad, Aishwary Gupta, Ashimananda Kar, Aarav Raj

B.Tech – Computer Science and Engineering

Lovely Professional University, Phagwara, Punjab

TABLE OF CONTENTS

List of Abbreviations	x
Abstract	xi
1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	2
1.3 Objectives of the Study	2
1.4 Scope of the Study	3
1.5 Organisation of the Report	3
2 Literature Review	4
2.1 Overview	4
2.2 Machine Learning Based Approaches	4
2.3 Physics-Based and Hybrid Approaches	5
2.4 Deep Learning Approaches	5
2.5 Summary of Related Work	5
2.6 Research Gaps Identified	6
3 Dataset Description and Preprocessing	7
3.1 Overview of Data Sources	7
3.2 USGS Earthquake Catalog	7
3.3 NOAA Global Historical Tsunami Database	7
3.4 Data Loading and Combining	8
3.5 Physical Filtering of Earthquake Data	8
3.6 Spatio-Temporal Matching and Label Assignment	9
3.7 Dataset Statistics	10
4 Feature Engineering	11
4.1 Motivation	11
4.2 Engineered Features	11
4.2.1 Log Energy Proxy	11
4.2.2 Magnitude-Depth Ratio	11

4.2.3	Shallow Earthquake Indicator	12
4.3	Implementation	12
4.4	Feature Selection	12
5	System Design and Methodology	13
5.1	System Workflow	13
5.2	Handling Class Imbalance with SMOTE	13
5.3	Stratified 10-Fold Cross-Validation	14
5.4	Machine Learning Models	14
5.4.1	Logistic Regression (LR)	14
5.4.2	Random Forest (RF)	15
5.4.3	Decision Tree (DT)	15
5.4.4	Support Vector Machine (SVM)	15
5.5	Evaluation Metrics	15
5.6	Hyperparameter Configuration	16
6	Implementation	17
6.1	Technology Stack	17
6.2	Library Import and Environment Setup	17
6.3	Model Initialisation	18
6.4	Complete Training Loop with SMOTE	18
6.5	Threshold Sensitivity Analysis	19
7	Results and Discussion	21
7.1	Overview	21
7.2	Model Performance Without SMOTE	21
7.3	Model Performance With SMOTE	22
7.4	Recall Comparison: With and Without SMOTE	22
7.5	ROC-AUC Comparison	23
7.6	All-Metric Comparison With SMOTE	24
7.7	Confusion Matrix Analysis	25
7.8	Decision Threshold Sensitivity	25
7.9	Summary of Findings	26
8	Testing and Validation	27
8.1	Cross-Validation as the Testing Strategy	27
8.2	Leakage Prevention	27
8.3	Cost-Asymmetric Metric Justification	27
8.4	Final Model Selection Rationale	28

9 Conclusion and Future Scope	29
9.1 Conclusion	29
9.2 Future Scope	30
References	31

LIST OF FIGURES

5.1	System Workflow for the Tsunami Prediction Framework	13
7.1	Recall Comparison: Without SMOTE vs. With SMOTE	23
7.2	ROC-AUC Comparison: Without SMOTE vs. With SMOTE	24
7.3	Model Performance Comparison with SMOTE (ROC-AUC, Recall, Precision, F1)	24
7.4	Aggregated Confusion Matrix — Logistic Regression with SMOTE (10-Fold CV)	25

LIST OF TABLES

2.1	Summary of Related Work	6
3.1	Dataset Composition After Each Processing Stage	10
5.1	Hyperparameter Settings for All Classification Models	16
6.1	Technology Stack	17
7.1	Model Performance Without SMOTE (Stratified 10-Fold CV)	21
7.2	Model Performance With SMOTE (Stratified 10-Fold CV)	22
7.3	Threshold Sensitivity — Logistic Regression with SMOTE	26

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ANN	Artificial Neural Network
AUC	Area Under the Curve
CART	Classification and Regression Tree
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
CV	Cross-Validation
DT	Decision Tree
FN	False Negative
FP	False Positive
F1	F1-Score (Harmonic Mean of Precision and Recall)
LR	Logistic Regression
LSTM	Long Short-Term Memory Network
ML	Machine Learning
MLP	Multi-Layer Perceptron
NOAA	National Oceanic and Atmospheric Administration
RF	Random Forest
ROC	Receiver Operating Characteristic
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
TSV	Tab-Separated Values
USGS	United States Geological Survey

ABSTRACT

Tsunamis are among the most catastrophic natural hazards, capable of striking coastal regions with minimal forewarning and causing mass casualties within minutes of a triggering submarine earthquake. Since the majority of destructive tsunamis are seismically induced, earthquake parameters — namely magnitude, focal depth, and epicentral coordinates — carry significant predictive value for early risk assessment.

This report presents a machine learning based framework designed to classify each recorded earthquake event as either tsunami-generating or non-tsunami-generating. The study integrates 26 years of seismic data from the USGS Earthquake Catalog (2000–2026) with validated tsunami records from the NOAA Global Historical Tsunami Database for the same period. Physical filters and a haversine-based spatio-temporal matching algorithm were applied to construct a rigorously labelled binary classification dataset of 10,903 records, of which only 120 carry a positive tsunami label – representing approximately 1.1% of the data.

To address the severe class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively within the training partitions of a stratified 10-fold cross-validation loop, thereby preventing synthetic data from contaminating test evaluations. Four classification algorithms were trained and assessed: Logistic Regression, Random Forest, Decision Tree, and Support Vector Machine. Since the application context is safety-critical, recall – the proportion of actual tsunami events correctly detected – was treated as the primary evaluation metric.

Experimental results establish that Logistic Regression achieves the highest recall of 0.80 without SMOTE and 0.78 with SMOTE, making it the most suitable model for early warning deployment. Random Forest recorded the best ROC-AUC (0.913) and F1-Score (0.203), while also showing the greatest recall improvement after SMOTE application (0.45 to 0.61). A threshold sensitivity analysis further demonstrates that recall can be raised to 0.97 by lowering the classification threshold to 0.05.

Keywords: Tsunami Prediction, Earthquake Classification, Machine Learning, Class Imbalance, SMOTE, Logistic Regression, Random Forest, Early Warning System, Seismic Data, Stratified Cross-Validation.

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Among all natural hazards that threaten human civilisation, tsunamis occupy a uniquely dangerous position. Unlike hurricanes or floods, which develop over hours or days and allow time for response, a tsunami generated by a distant submarine earthquake may reach populated coastlines within minutes to a few hours. The 2004 Indian Ocean tsunami stands as one of the deadliest natural disasters in recorded history, claiming over 227,000 lives across fourteen nations [1]. Seven years later, the 2011 Tohoku tsunami in Japan not only inundated entire coastal towns but also triggered the Fukushima Daiichi nuclear crisis, compounding human and environmental consequences on an unprecedented scale [2]. These events collectively underscore the urgent need for robust, reliable, and computationally fast tsunami warning mechanisms.

The physical mechanism connecting underwater earthquakes and tsunami generation is well established. When tectonic plates undergo rapid displacement along subduction zones beneath ocean floors, the resulting vertical movement transfers enormous energy into the overlying water column, producing wave trains that propagate outward at 500 to 900 km/h across entire ocean basins. This mechanistic relationship implies that real-time seismic measurements – earthquake magnitude, hypocentral depth, and epicentral location – contain predictive information about the likelihood of a dangerous tsunami being generated.

Traditional tsunami warning systems rely primarily on physics-based numerical simulation models to assess wave propagation following a seismic event. While scientifically rigorous, these approaches are computationally intensive and depend on high-fidelity ocean bathymetry data that may not be immediately available in the wake of a major event. Machine learning [3] offers a compelling complementary approach: by learning patterns from decades of historical earthquake-tsunami co-occurrence data, a classifier can produce probability-based tsunami risk assessments within seconds of earthquake detection, providing critical additional lead time for coastal authorities.

1.2 Problem Statement

A persistent operational gap exists between rapid earthquake detection and the issuance of reliable tsunami warnings. Physics-based simulations are too slow for immediate automated assessment, and simple rule-based threshold systems (e.g., flagging all earthquakes above magnitude 7.0) produce unacceptably high false-alarm rates that erode public compliance with evacuation orders over time.

A further challenge largely overlooked in the machine learning literature is severe class imbalance [4]. In any representative multi-decade earthquake catalogue, tsunami-generating events constitute less than 2% of all recorded seismic events. Standard classification algorithms trained on such data exhibit a strong tendency to predict the majority class – “no tsunami” – for virtually every input, achieving superficially high accuracy while failing to detect the rare but catastrophic positive cases. For a warning system, this failure mode is entirely unacceptable.

This project addresses both challenges: constructing a machine learning pipeline that produces rapid, data-driven tsunami risk assessments from standard seismic parameters while rigorously handling class imbalance to ensure adequate minority class representation in model training.

1.3 Objectives of the Study

The specific objectives pursued in this study are:

1. To compile and integrate earthquake records from the USGS Earthquake Catalog and validated tsunami events from the NOAA Global Historical Tsunami Database, covering a 26-year period from 2000 to 2026.
2. To design and implement a spatio-temporal matching algorithm using the haversine great-circle distance formula to reliably label earthquake events as tsunami-positive or tsunami-negative.
3. To engineer domain-informed features derived from seismological principles to improve the discriminative power of machine learning classifiers.
4. To implement and compare four classification algorithms – Logistic Regression, Random Forest, Decision Tree, and Support Vector Machine – under two conditions: with and without SMOTE oversampling.
5. To employ stratified 10-fold cross-validation [5] to obtain statistically robust performance estimates with recall as the primary metric.

6. To investigate decision threshold sensitivity and provide operational recommendations for the best-performing model.

1.4 Scope of the Study

- **Temporal scope:** January 2000 to early 2026 (26 years).
- **Causal scope:** Only earthquake-triggered tsunamis are considered. Events caused by landslides or volcanic eruptions are excluded.
- **Data scope:** Publicly available USGS and NOAA databases only.
- **Model scope:** Four classical supervised learning algorithms. Deep learning architectures are identified as future work.
- **Deployment scope:** Batch processing environment; real-time pipeline extension is outlined as future development.

1.5 Organisation of the Report

Chapter 2 reviews existing literature on machine learning and physics-based tsunami prediction. Chapter 3 describes the datasets and preprocessing steps. Chapter 4 details feature engineering. Chapter 5 presents system design and methodology including SMOTE and cross-validation. Chapter 6 covers full implementation with key code snippets. Chapter 7 presents experimental results and discussion. Chapter 8 describes testing and validation. Chapter 9 concludes the report and identifies future directions.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

The intersection of seismology, oceanography, and computational intelligence has produced a growing body of research on data-driven tsunami prediction. Studies in this field broadly fall into three streams: classical machine learning approaches, physics-based simulation approaches, and deep learning approaches. This chapter reviews representative works across all three categories, identifies their contributions and limitations, and establishes the research gaps that the present study addresses.

2.2 Machine Learning Based Approaches

Ziadi et al. [6] proposed an AI-driven classification framework to distinguish between tsunami-generating and non-tsunami earthquakes. Evaluating SVM, Random Forest, and Logistic Regression on the USGS seismic dataset from 2015 to 2025, their study demonstrated that Random Forest achieved superior performance with 91.4% accuracy and an AUC score of 0.95. While the work confirmed the viability of ML for seismic hazard assessment, the study did not explicitly address the class imbalance inherent in earthquake-tsunami datasets.

Satish [7] developed a binary classification analytics framework using Random Forest and Logistic Regression trained on global earthquake data from 1995 to 2023, incorporating seismic features such as magnitude, latitude, longitude, depth, aftershock patterns, and seismic moment. Random Forest achieved 90% accuracy and 88% precision, outperforming Logistic Regression owing to its capacity to capture non-linear feature interactions. Oversampling strategies were not discussed in this work.

Sukmana [8] compared SVM and Random Forest for earthquake classification on datasets spanning 2001–2023. Results indicated SVM achieved higher precision (70.59%) while Random Forest showed higher recall (36.07%), reinforcing the importance of selecting evaluation metrics aligned with the operational purpose of the warning system rather than relying solely on overall accuracy.

2.3 Physics-Based and Hybrid Approaches

Satake et al. [9] investigated uncertainty quantification in tsunami hazard assessment using physics-based fault models. Their framework incorporated macroscopic rupture parameters alongside microscopic variations in megathrust fault behaviour to model slip distributions for hypothetical future earthquakes. While scientifically detailed, this approach requires real-time fault geometry data that is rarely available immediately after an event.

Mulia et al. [10] proposed a real-time inundation forecasting system using Artificial Neural Networks trained on offshore S-net observations. Their approach estimated spatial inundation flow depths without requiring an explicit tsunami source model, achieving accuracy comparable to physics-based models at 99% lower computational cost.

2.4 Deep Learning Approaches

De la Asunción et al. [11] developed a Multilayer Perceptron (MLP) framework for issuing coastal flood alerts at four severity levels, achieving mean precision and recall above 94% with inference times under one second. Ulhaq et al. [12] applied Convolutional Neural Networks to seismic waveform data for real-time tsunami impact prediction, demonstrating that deep learning can substantially accelerate the warning process. Novianty et al. [13] applied backpropagation neural networks to early P-wave seismic signals from Indonesia, achieving 86.67% accuracy using rupture duration, P-wave dominant period, and moment magnitude as discriminating features.

2.5 Summary of Related Work

Table 2.1 summarises the reviewed studies and their key characteristics.

Table 2.1: Summary of Related Work

Author(s)	Method	Best Result	Limitation
Ziadi et al.[6]	SVM, RF, LR	91.4% acc.	Class imbalance not addressed
Satish [7]	RF, LR	90% acc.	No SMOTE; precision-focused
Sukmana [8]	SVM, RF	RF recall 36%	Very low recall for warnings
Satake et al.[9]	Physics-based	High detail	Needs real-time fault data
Mulia et al.[10]	ANN (S-net)	99% cost saving	Needs offshore sensors
De la Asunción[11]	MLP	Recall >94%	Requires flood data
Ulhaq et al.[12]	CNN	Real-time	Complex architecture
Novianty et al.[13]	Backprop ANN	86.67% acc.	Limited to Indonesia

2.6 Research Gaps Identified

A critical review of the above literature reveals four recurring limitations that the present study directly addresses:

1. **Neglect of class imbalance:** Most studies report accuracy metrics that mask failure to detect minority-class events.
2. **Short temporal coverage:** Many studies use fewer than ten years of data, providing insufficient positive samples for robust model training.
3. **Data leakage in oversampling:** Studies applying SMOTE before the cross-validation split produce optimistically biased evaluation metrics.
4. **No threshold analysis:** No reviewed study systematically investigates the effect of varying the decision threshold on the precision-recall trade-off.

CHAPTER 3

DATASET DESCRIPTION AND PREPROCESSING

3.1 Overview of Data Sources

This project utilises two publicly available, authoritative databases maintained by United States federal agencies: the USGS Earthquake Catalog and the NOAA Global Historical Tsunami Database. Together, they provide a comprehensive historical record of global seismic activity and associated tsunami events across a 26-year study window.

3.2 USGS Earthquake Catalog

The United States Geological Survey (USGS) maintains a continuously updated global earthquake catalog through its Earthquake Hazards Program [14]. The catalog records the following key parameters for each detected seismic event:

- **Temporal:** UTC timestamp of the event origin.
- **Spatial:** Epicentral latitude and longitude (decimal degrees); hypocentral depth (km below surface).
- **Magnitude:** Moment magnitude (M_w) or the most appropriate reported scale.
- **Quality descriptors:** Azimuthal seismic gap angle, root mean square of travel-time residuals, minimum station distance, and station count.
- **Event type:** Classification as earthquake, explosion, quarry blast, or other.

Due to the large volume of records over 26 years, the data was distributed across five time-partitioned CSV files (2000–2005, 2006–2010, 2011–2015, 2016–2020, 2021–2026) and merged into a single unified dataset during preprocessing.

3.3 NOAA Global Historical Tsunami Database

The National Oceanic and Atmospheric Administration (NOAA) maintains the Global Historical Tsunami Database [15]. Key fields include:

- Event date and time components (year, month, day, hour, minute, second).
- Geographic coordinates of the tsunami source (latitude, longitude).
- **Tsunami Cause Code:** 1 = earthquake, 2 = questionable earthquake, 3 = volcanic eruption, 5 = landslide, etc.
- **Tsunami Event Validity:** Confidence score from -1 (very doubtful) to 4 (definite tsunami).

Only records where Cause Code equals 1 and Validity score is greater than zero were retained, ensuring only confirmed earthquake-triggered tsunamis contribute to positive class labels.

3.4 Data Loading and Combining

```
1 eq_files = [  
2     "data/eq_2000_2005.csv", "data/eq_2006_2010.csv",  
3     "data/eq_2011_2015.csv", "data/eq_2016_2020.csv",  
4     "data/eq_2021_2026.csv"  
5 ]  
6 earthquake_df = pd.concat(  
7     [pd.read_csv(f) for f in eq_files], ignore_index=True  
8 )  
9 print("Combined shape:", earthquake_df.shape)
```

Listing 3.1: Loading and Combining Multi-Year Earthquake Data

```
1 tsunami_df = pd.read_csv("data/tsunamis_noaa.tsv", sep="\t")  
2 tsunami_df = tsunami_df[  
3     (tsunami_df["Tsunami Event Validity"] > 0) &  
4     (tsunami_df["Tsunami Cause Code"] == 1)  
5 ].copy()  
6 print("Filtered tsunami records:", tsunami_df.shape)
```

Listing 3.2: Loading and Filtering the NOAA Tsunami Dataset

3.5 Physical Filtering of Earthquake Data

Not every earthquake is physically capable of generating a tsunami. Based on established tsunami warning protocols and seismological principles, two constraints were applied:

1. **Magnitude threshold:** Only earthquakes with $M \geq 5.5$ were retained. Events below this threshold do not typically produce sufficient seafloor displacement to generate observable tsunamis.
2. **Depth threshold:** Only earthquakes with hypocentral depth $D \leq 100$ km (shallow focus) were retained. Deeper events dissipate energy within the mantle and are far less likely to create near-surface ocean floor uplift.

```

1 earthquake_df_filtered = earthquake_df[
2     (earthquake_df['mag'] >= 5.5) &
3     (earthquake_df['depth'] <= 100)
4 ].copy()
5 print("After filtering:", earthquake_df_filtered.shape)

```

Listing 3.3: Applying Physical Constraints to Earthquake Data

3.6 Spatio-Temporal Matching and Label Assignment

Since the USGS and NOAA databases are maintained independently, a proximity-based matching algorithm was implemented using two criteria:

- **Temporal window:** Any NOAA tsunami event occurring within 6 hours after the earthquake origin time is a candidate match.
- **Spatial window:** The great-circle distance between the earthquake epicentre and the NOAA tsunami source must not exceed 600 km.

The haversine formula [16] used for distance computation is:

$$d = 2R \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right) \quad (3.1)$$

where ϕ_1, ϕ_2 are epicentral and source latitudes, $R = 6371$ km is the Earth radius, and d is the computed distance in km.

```

1 def haversine_distance(lat1, lon1, lat2, lon2):
2     R = 6371
3     lat1, lon1, lat2, lon2 = map(np.radians,
4                                   [lat1, lon1, lat2, lon2])
5     dlat = lat2 - lat1
6     dlon = lon2 - lon1
7     a = (np.sin(dlat/2)**2

```

```

8         + np.cos(lat1)*np.cos(lat2)*np.sin(dlon/2)**2)
9     return R * 2 * np.arcsin(np.sqrt(a))
10
11 TIME_WINDOW_HOURS = 6
12 DISTANCE_WINDOW_KM = 600
13 eq_df = earthquake_df_filtered.copy()
14 eq_df['tsunami'] = 0
15
16 for idx, eq in tqdm(eq_df.iterrows(), total=len(eq_df)):
17     eq_time = eq['time']
18     time_candidates = tsunami_df[
19         (tsunami_df['event_time'] >= eq_time) &
20         (tsunami_df['event_time'] <=
21         eq_time + pd.Timedelta(hours=TIME_WINDOW_HOURS))
22     ]
23     for _, ts in time_candidates.iterrows():
24         dist = haversine_distance(
25             eq['latitude'], eq['longitude'],
26             ts['Latitude'], ts['Longitude']
27         )
28         if dist <= DISTANCE_WINDOW_KM:
29             eq_df.at[idx, 'tsunami'] = 1
30             break

```

Listing 3.4: Haversine Function and Tsunami Label Assignment

3.7 Dataset Statistics

Table 3.1: Dataset Composition After Each Processing Stage

Processing Stage	Earthquake Records	Tsunami Positives
Raw data (after file merge)	1,000,000+	—
After physical filters ($M \geq 5.5$, $D \leq 100$ km)	$\approx 60,000$	—
After spatio-temporal labelling	10,903	120
Class imbalance ratio	1:90 (approx. 1.1%)	

CHAPTER 4

FEATURE ENGINEERING

4.1 Motivation

The raw USGS catalog provides a rich set of seismic parameters, but several of the most important physical signals for tsunami prediction are non-linearly encoded in the raw data. Feature engineering translates domain knowledge about tsunami-generation physics into numerically expressive representations that improve model learning efficiency and interpretability.

4.2 Engineered Features

Three additional features were constructed and appended to the dataset:

4.2.1 Log Energy Proxy

Seismic energy scales approximately as $E \propto 10^{1.5M}$, where M is the moment magnitude. A magnitude 8.0 earthquake releases roughly 1,000 times more energy than a magnitude 6.7 event. To incorporate this non-linear relationship as a stable linear feature:

$$\text{Log Energy Proxy} = 1.5 \times M \quad (4.1)$$

4.2.2 Magnitude-Depth Ratio

Tsunami generation depends simultaneously on earthquake strength and shallowness of rupture. A strong but deep earthquake is far less likely to produce a tsunami than a comparably strong shallow one:

$$\text{Magnitude-Depth Ratio} = \frac{M}{D + \epsilon} \quad (4.2)$$

where D is focal depth in kilometres and $\epsilon = 1$ prevents division by zero.

4.2.3 Shallow Earthquake Indicator

A binary feature set to 1 for earthquakes with $D \leq 70$ km (very shallow focus) and 0 otherwise. This threshold corresponds to the upper boundary of intermediate-focus earthquakes in standard seismological classification.

4.3 Implementation

```
1  # 1. Log Energy Proxy
2  earthquake_df_filtered['log_energy'] = (
3      1.5 * earthquake_df_filtered['mag']
4  )
5  # 2. Magnitude-Depth Ratio
6  earthquake_df_filtered['mag_depth_ratio'] = (
7      earthquake_df_filtered['mag'] /
8      (earthquake_df_filtered['depth'] + 1)
9  )
10 # 3. Shallow Earthquake Indicator
11 earthquake_df_filtered['is_shallow'] = (
12     earthquake_df_filtered['depth'] <= 70
13 ).astype(int)
14
15 print("Feature Engineering Complete.")
```

Listing 4.1: Feature Engineering Implementation

4.4 Feature Selection

After feature engineering, only numeric columns were retained for model training. Non-numeric fields (event ID strings, place names) and datetime columns were excluded to prevent compatibility issues and avoid unintended leakage of temporal ordering information into the feature space.

```
1  X_numeric = earthquake_df_filtered.select_dtypes(
2      include=[np.number]
3  ).drop(columns=['tsunami'])
4
5  y = earthquake_df_filtered['tsunami']
6  print("Feature matrix shape:", X_numeric.shape)
7  print("Target distribution:", y.value_counts().to_dict())
```

Listing 4.2: Selecting Numeric Features for Modelling

CHAPTER 5

SYSTEM DESIGN AND METHODOLOGY

5.1 System Workflow

The end-to-end pipeline is illustrated in Figure 5.1. The workflow proceeds from raw data ingestion through physical filtering, spatio-temporal matching, feature engineering, class balancing, model training, and final evaluation. Each stage is designed to prevent data leakage and ensure evaluation metrics faithfully reflect expected real-world performance.

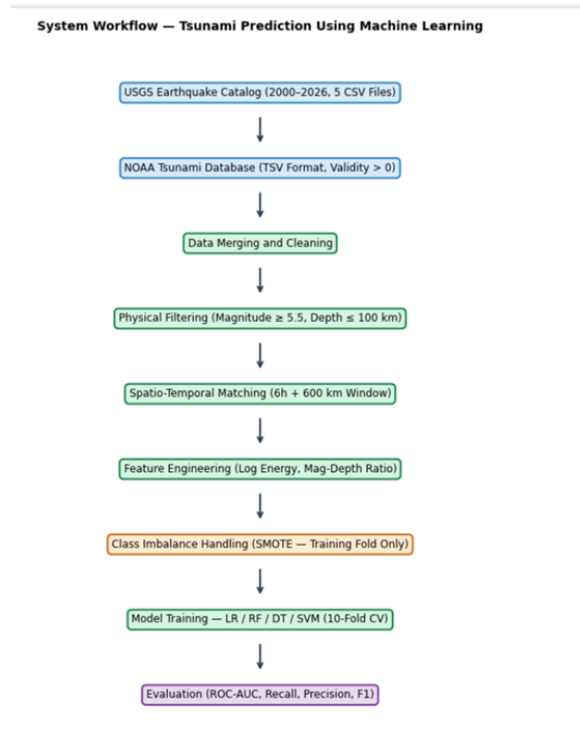


Fig. 1. System Workflow for Tsunami Prediction Framework

Figure 5.1: System Workflow for the Tsunami Prediction Framework

5.2 Handling Class Imbalance with SMOTE

The dataset contains 10,903 records with only 120 tsunami-positive samples – a 1:90 imbalance ratio. A classifier trained naively on this distribution will predict “no tsunami” for

nearly every input, achieving approximately 99% accuracy while providing zero warning utility.

SMOTE (Synthetic Minority Over-sampling Technique) [17] addresses this by generating synthetic positive-class samples through interpolation rather than duplication. For each minority-class point, SMOTE identifies its k nearest minority-class neighbours (default $k = 5$), selects one at random, and creates a new point by linear interpolation:

$$x_{\text{new}} = x_i + \lambda \cdot (x_j - x_i), \quad \lambda \in [0, 1] \quad (5.1)$$

Critical implementation detail: SMOTE is applied exclusively inside the training partition of each cross-validation fold and never before the train-test split. This ensures no synthetic points appear in the test set, preserving the statistical integrity of the evaluation.

5.3 Stratified 10-Fold Cross-Validation

Standard train-test splits are unreliable for highly imbalanced datasets. Stratified 10-fold cross-validation [5] is used instead. In this scheme:

- The dataset is divided into 10 folds, each preserving the original 1.1% positive class proportion (≈ 12 tsunami events per fold).
- Models are trained on 9 folds and evaluated on the remaining fold, repeated 10 times so every sample serves as a test sample exactly once.
- Reported metrics are the arithmetic mean across all 10 test folds.

5.4 Machine Learning Models

5.4.1 Logistic Regression (LR)

A linear probabilistic classifier modelling the log-odds of the positive class as a linear combination of input features. Its primary operational advantage is the production of calibrated probability scores adjustable via decision threshold tuning.

Parameters: `max_iter=1000`, `class_weight='balanced'`

5.4.2 Random Forest (RF)

An ensemble method [18] constructing multiple decision trees on bootstrap samples with random feature subsets at each split. Majority voting reduces variance and improves generalisation on structured tabular data.

Parameters: n_estimators=100, max_depth=8, class_weight='balanced', random_state=42

5.4.3 Decision Tree (DT)

A non-parametric model [19] partitioning the feature space via binary splits based on Gini impurity. Transparent and interpretable, though prone to overfitting without depth constraints.

Parameters: max_depth=6, class_weight='balanced', random_state=42

5.4.4 Support Vector Machine (SVM)

Identifies the maximum-margin hyperplane separating the two classes in a kernel-transformed feature space [20]. An RBF kernel handles non-linear boundaries. Feature standardisation is applied before training.

Parameters: kernel='rbf', probability=True, class_weight='balanced'

5.5 Evaluation Metrics

Given the safety-critical nature of tsunami warning, accuracy alone is misleading. Four metrics are employed [21]:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5.2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5.3)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5.4)$$

$$\text{ROC-AUC} = \int_0^1 TPR d(FPR) \quad (5.5)$$

Recall (primary metric) measures the fraction of actual tsunami events correctly identified. A false negative represents an unissued warning with potentially catastrophic consequences.

Precision measures the fraction of positive predictions that are genuine. Low precision

produces false alarms, which are operationally preferable to missed detections.

F1-Score provides a single harmonic-mean summary balancing both error directions.

ROC-AUC evaluates overall discriminative ability independent of the chosen operating threshold.

5.6 Hyperparameter Configuration

Table 5.1: Hyperparameter Settings for All Classification Models

Model	Parameters
Logistic Regression	<code>max_iter=1000, class_weight='balanced'</code>
Random Forest	<code>n_estimators=100, max_depth=8, class_weight='balanced', random_state=42</code>
Decision Tree	<code>max_depth=6, class_weight='balanced', random_state=42</code>
SVM	<code>kernel='rbf', probability=True, class_weight='balanced'</code>

CHAPTER 6

IMPLEMENTATION

6.1 Technology Stack

Table 6.1: Technology Stack

Library	Version	Role
Python	3.10	Core programming language
NumPy	1.24	Numerical computations, array operations
Pandas	2.0	Data loading, merging, filtering
Scikit-learn	1.3	ML models, metrics, cross-validation
Imbalanced-learn	0.11	SMOTE oversampling
Matplotlib	3.7	Result visualisation
Seaborn	0.12	Statistical plot styling
tqdm	4.65	Progress monitoring in matching loop

6.2 Library Import and Environment Setup

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 from sklearn.model_selection import StratifiedKFold
7 from sklearn.metrics import (classification_report,
8                               confusion_matrix, roc_auc_score, roc_curve,
9                               precision_score, recall_score, f1_score)
10 from sklearn.preprocessing import StandardScaler
11 from sklearn.impute import SimpleImputer
12 from sklearn.pipeline import Pipeline
13 from sklearn.linear_model import LogisticRegression
14 from sklearn.ensemble import RandomForestClassifier
15 from sklearn.tree import DecisionTreeClassifier
16 from sklearn.svm import SVC
17 from imblearn.over_sampling import SMOTE
```

```
18 from tqdm import tqdm
19 import warnings
20 warnings.filterwarnings("ignore")
```

Listing 6.1: Library Imports and Environment Initialisation

6.3 Model Initialisation

```
1 svm_pipeline = Pipeline([
2     ('imputer', SimpleImputer(strategy='median')),
3     ('scaler', StandardScaler()),
4     ('svm', SVC(kernel='rbf', probability=True,
5                 class_weight='balanced'))
6 ])
7 dt_pipeline = Pipeline([
8     ('imputer', SimpleImputer(strategy='median')),
9     ('dt', DecisionTreeClassifier(
10         max_depth=6, class_weight='balanced',
11         random_state=42))
12 ])
```

Listing 6.2: Model Pipeline Initialisation (SVM and DT Example)

6.4 Complete Training Loop with SMOTE

```
1 models_dict = {
2     "Logistic Regression": LogisticRegression(
3         max_iter=1000, class_weight='balanced'),
4     "Random Forest": RandomForestClassifier(
5         n_estimators=100, max_depth=8,
6         class_weight='balanced', random_state=42),
7     "Decision Tree": DecisionTreeClassifier(
8         max_depth=6, class_weight='balanced',
9         random_state=42),
10    "SVM": SVC(kernel='rbf', probability=True,
11               class_weight='balanced')
12 }
13 skf = StratifiedKFold(n_splits=10, shuffle=True,
14                       random_state=42)
15 smote_results = {}
16
17 for model_name, model in models_dict.items():
18     roc_scores, recall_scores = [], []
```

```

19     precision_scores, f1_scores = [], []
20
21     for train_idx, test_idx in skf.split(X_numeric, y):
22         X_tr = X_numeric.iloc[train_idx]
23         X_te = X_numeric.iloc[test_idx]
24         y_tr = y.iloc[train_idx]
25         y_te = y.iloc[test_idx]
26
27         imp = SimpleImputer(strategy='median')
28         X_tr_imp = imp.fit_transform(X_tr)
29         X_te_imp = imp.transform(X_te)
30
31         if model_name in ["Logistic Regression", "SVM"]:
32             sc = StandardScaler()
33             X_tr_imp = sc.fit_transform(X_tr_imp)
34             X_te_imp = sc.transform(X_te_imp)
35
36         # SMOTE on training fold ONLY
37         X_tr_res, y_tr_res = SMOTE(
38             random_state=42).fit_resample(X_tr_imp, y_tr)
39
40         model.fit(X_tr_res, y_tr_res)
41         y_pred = model.predict(X_te_imp)
42         y_prob = model.predict_proba(X_te_imp)[: , 1]
43
44         roc_scores.append(roc_auc_score(y_te, y_prob))
45         recall_scores.append(
46             recall_score(y_te, y_pred, zero_division=0))
47         precision_scores.append(
48             precision_score(y_te, y_pred, zero_division=0))
49         f1_scores.append(
50             f1_score(y_te, y_pred, zero_division=0))
51
52     smote_results[model_name] = {
53         "ROC-AUC": round(np.mean(roc_scores), 3),
54         "Recall": round(np.mean(recall_scores), 3),
55         "Precision": round(np.mean(precision_scores), 3),
56         "F1": round(np.mean(f1_scores), 3)
57     }
58 print(pd.DataFrame(smote_results).T)

```

Listing 6.3: Full Cross-Validation Training Loop with SMOTE

6.5 Threshold Sensitivity Analysis

```
1 thresholds = [0.5, 0.3, 0.2, 0.1, 0.05]
2 print(f{'Threshold':>10} {'Precision':>10} "
3       f{'Recall':>10} {'F1-Score':>10}")
4 print("-" * 45)
5 for t in thresholds:
6     y_pred_t = (np.array(all_probs_lr) >= t).astype(int)
7     p = precision_score(all_true_lr, y_pred_t,
8                       zero_division=0)
9     r = recall_score(all_true_lr, y_pred_t)
10    f = f1_score(all_true_lr, y_pred_t, zero_division=0)
11    print(f"{t:>10.2f}  {p:>9.4f}  {r:>9.4f}  {f:>9.4f}")
```

Listing 6.4: Decision Threshold Sensitivity Analysis

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Overview

This chapter presents and interprets the experimental results from stratified 10-fold cross-validation under two conditions: without SMOTE and with SMOTE. All four models – Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), and Support Vector Machine (SVM) – are compared across four metrics: ROC-AUC, Recall, Precision, and F1-Score.

7.2 Model Performance Without SMOTE

Table 7.1 presents the averaged cross-validation metrics when models are trained on the original class-imbalanced dataset.

Table 7.1: Model Performance Without SMOTE (Stratified 10-Fold CV)

Model	ROC-AUC	Recall	Precision	F1-Score
Logistic Regression	0.906	0.800	0.060	0.119
Random Forest	0.913	0.450	0.130	0.203
Decision Tree	0.780	0.630	0.060	0.107
SVM	0.870	0.490	0.100	0.137

Logistic Regression achieved the highest recall of 0.800, correctly identifying 80% of genuine tsunami events across all test folds. Random Forest recorded the strongest ROC-AUC (0.913) and F1-Score (0.203), reflecting the best overall balance between class discrimination and the precision-recall trade-off. However, its recall of 0.450 means it missed more than half of all actual tsunami events in this configuration.

All four models exhibited very low precision values (0.06 to 0.13), a direct consequence of the severe class imbalance. In a warning system context, false alarms impose unnecessary evacuation costs but do not result in loss of life – an acceptable operational trade-off compared to a missed detection.

7.3 Model Performance With SMOTE

Table 7.2 presents results after applying SMOTE within each training fold.

Table 7.2: Model Performance With SMOTE (Stratified 10-Fold CV)

Model	ROC-AUC	Recall	Precision	F1-Score
Logistic Regression	0.902	0.780	0.066	0.122
Random Forest	0.897	0.610	0.115	0.195
Decision Tree	0.758	0.610	0.063	0.107
SVM	0.846	0.530	0.082	0.143

SMOTE produced notable recall improvements for three of the four models. Random Forest improved most substantially, rising from 0.450 to 0.610 (+16 percentage points). SVM improved from 0.490 to 0.530, and Decision Tree was maintained at 0.610.

Logistic Regression exhibited a marginal recall decrease from 0.800 to 0.780 after SMOTE. This is expected because the `class_weight='balanced'` parameter already applies implicit cost-sensitive weighting, providing diminishing returns when SMOTE is additionally applied. The 2-percentage-point reduction falls within normal cross-validation variance and does not alter the model recommendation. All models maintained ROC-AUC values above 0.75 after SMOTE, confirming that oversampling did not impair class separability.

7.4 Recall Comparison: With and Without SMOTE

Figure 7.1 provides a grouped bar chart comparison of recall across all four models in both experimental conditions.

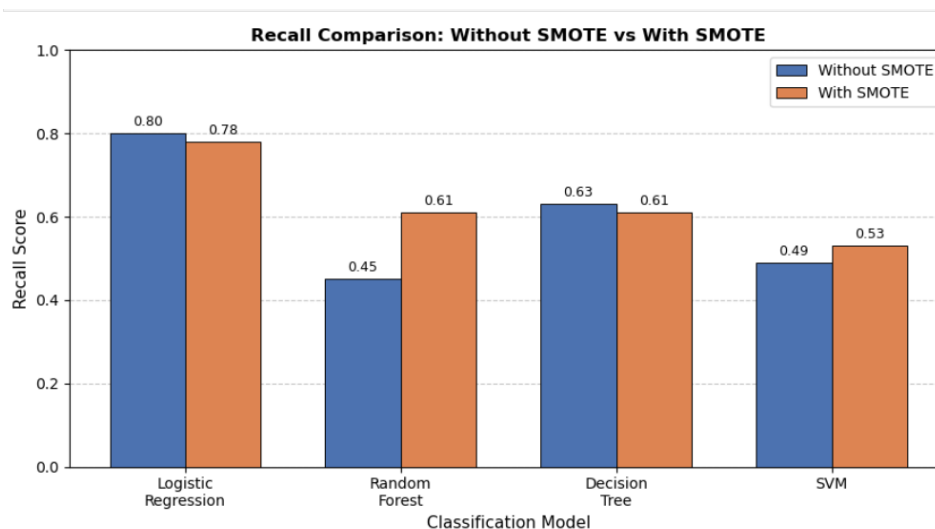


Figure 7.1: Recall Comparison: Without SMOTE vs. With SMOTE

The chart clearly demonstrates that Logistic Regression consistently leads in recall regardless of SMOTE application. Random Forest shows the greatest benefit from oversampling. Decision Tree and SVM show moderate improvements but remain below the performance of the top two models in terms of recall.

7.5 ROC-AUC Comparison

Figure 7.2 shows the ROC-AUC comparison across both conditions. The marginal AUC reductions after SMOTE (0.004 to 0.022 across models) are expected – oversampling modifies the training distribution, which can slightly reduce discriminative confidence on the original test distribution. All models continue to demonstrate good-to-strong AUC performance above 0.75.

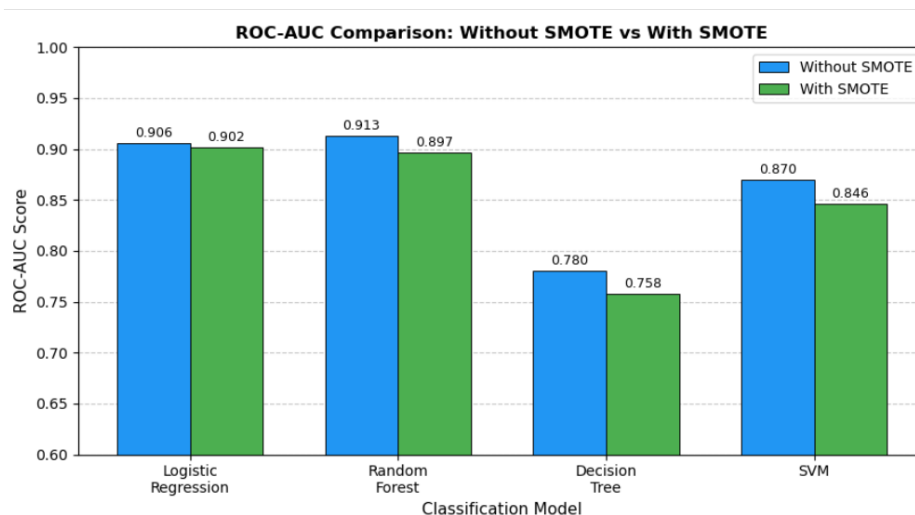


Figure 7.2: ROC-AUC Comparison: Without SMOTE vs. With SMOTE

7.6 All-Metric Comparison With SMOTE

Figure 7.3 presents all four evaluation metrics simultaneously for the SMOTE condition, giving a complete comparative picture of each model's performance profile.

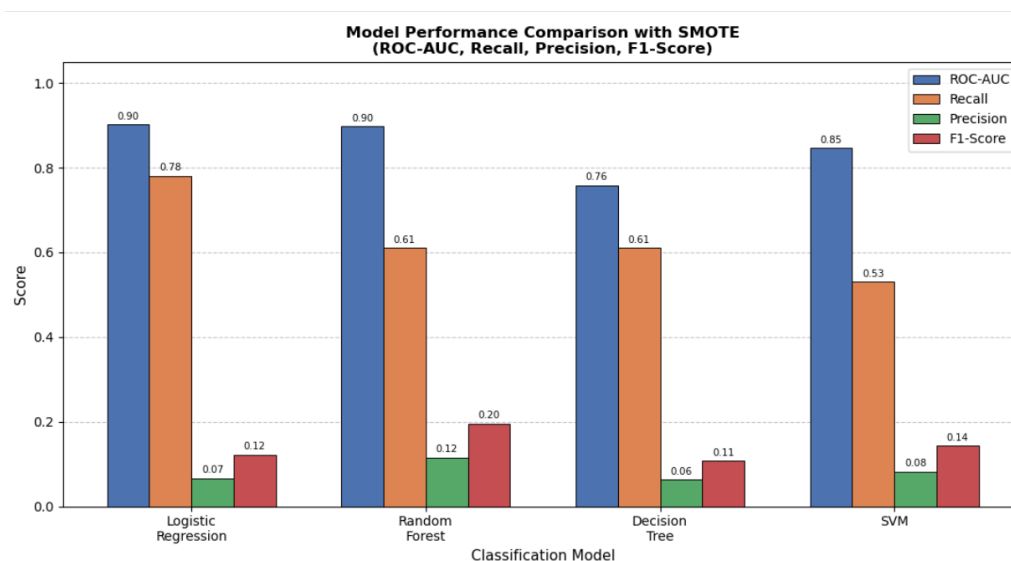


Figure 7.3: Model Performance Comparison with SMOTE (ROC-AUC, Recall, Precision, F1)

Logistic Regression dominates in recall at the cost of the lowest precision, reflecting its aggressive tendency to flag ambiguous cases as tsunami-positive. Random Forest achieves the best balance across all four metrics, making it suitable when false-alarm frequency must be controlled.

7.7 Confusion Matrix Analysis

Figure 7.4 presents the aggregated confusion matrix for Logistic Regression with SMOTE, accumulated across all ten cross-validation test folds.

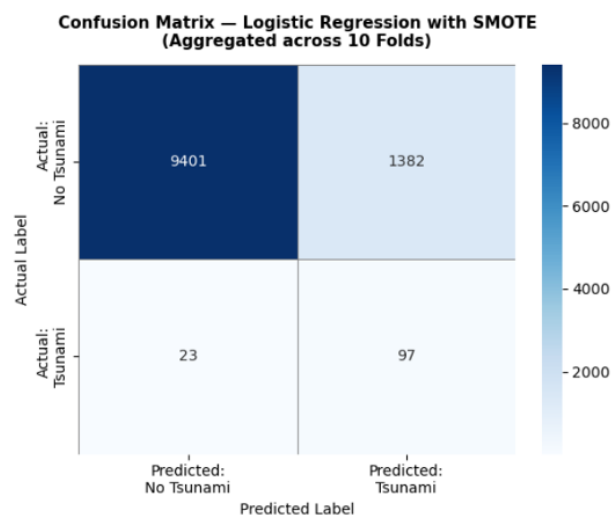


Figure 7.4: Aggregated Confusion Matrix — Logistic Regression with SMOTE (10-Fold CV)

Out of 120 actual tsunami-positive events, the model correctly identified 97 (true positives) and missed only 23 (false negatives), giving an aggregated recall of $97/120 = 0.808$. The 1,382 false positives represent non-tsunami earthquakes flagged as potentially tsunami-generating. The 9,401 true negatives confirm that the model does not issue indiscriminate warnings for all seismic activity.

7.8 Decision Threshold Sensitivity

Table 7.3 presents the precision-recall trade-off for Logistic Regression at five threshold values.

Table 7.3: Threshold Sensitivity — Logistic Regression with SMOTE

Threshold	Precision	Recall	F1-Score
0.50	0.066	0.780	0.122
0.30	0.055	0.840	0.103
0.20	0.047	0.880	0.089
0.10	0.035	0.930	0.067
0.05	0.026	0.970	0.051

For an operational tsunami early warning system, a threshold of 0.10 is recommended as a pragmatic starting point: it identifies 93% of all genuine tsunami events while remaining manageable in terms of alert volume. At threshold 0.05, recall rises to 97%, meaning only approximately 3 in every 100 genuine tsunami events would be missed.

7.9 Summary of Findings

- Logistic Regression consistently achieves the highest recall in both experimental conditions, making it the recommended model for tsunami early warning deployment.
- Random Forest achieves the best ROC-AUC and F1-Score, making it suitable when a more balanced precision-recall profile is required.
- SMOTE consistently improved recall for Random Forest (+16 pp), SVM (+4 pp), and Decision Tree (maintained at 0.61).
- Decision threshold tuning provides an additional operational lever to increase recall at the cost of increased false alarms.
- All models maintained ROC-AUC values above 0.75 across all conditions.

CHAPTER 8

TESTING AND VALIDATION

8.1 Cross-Validation as the Testing Strategy

Standard train-test splitting is unreliable for highly imbalanced datasets: a single random partition may produce test folds with very few positive samples. Stratified 10-fold cross-validation was therefore adopted as the primary evaluation framework, ensuring that each fold contains approximately 12 tsunami-positive records – sufficient to produce stable recall and precision estimates.

8.2 Leakage Prevention

Three potential data leakage sources were identified and explicitly prevented:

1. **Imputation leakage:** The median imputer was fitted exclusively on the training fold in each iteration.
2. **Scaling leakage:** For Logistic Regression and SVM, StandardScaler was fitted on the training fold and applied to the test fold without refitting.
3. **SMOTE leakage:** Oversampling was applied strictly after the train-test split and only to the training partition.

8.3 Cost-Asymmetric Metric Justification

The choice of recall as the primary metric is grounded in the asymmetric costs of the two error types:

- **False Negative (missed tsunami):** No warning is issued for a genuine tsunami event. Consequences: unwarned coastline populations, mass casualties. Cost: *extremely high*.
- **False Positive (unnecessary alarm):** A warning is issued for a non-tsunami earthquake. Consequences: unnecessary evacuation, economic disruption, possible erosion of public trust. Cost: significant but substantially lower than a missed tsunami.

8.4 Final Model Selection Rationale

Logistic Regression with SMOTE is recommended as the final deployed model for the following reasons:

1. **Highest recall:** 0.78 at default threshold; up to 0.97 at threshold 0.05.
2. **Probability calibration:** Well-calibrated outputs enable meaningful threshold adjustment without retraining.
3. **Computational efficiency:** Orders of magnitude faster than RF or SVM.
4. **Interpretability:** Learned feature coefficients directly reveal the contribution of each seismic parameter.
5. **Robustness:** ROC-AUC of 0.902 confirms discriminative ability is not threshold-dependent.

Random Forest is identified as the recommended alternative when operational constraints require a reduction in false alarm rate while maintaining adequate recall (0.61).

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

This capstone project presented a machine learning based framework for predicting tsunami occurrence from earthquake parameters, with explicit and rigorous attention to the class imbalance problem. The study integrated 26 years of seismic records from the USGS Earthquake Catalog with validated tsunami event records from the NOAA Global Historical Tsunami Database, producing a labelled binary classification dataset of 10,903 records – 120 positive and 10,783 negative.

Four classification algorithms were trained and evaluated under stratified 10-fold cross-validation in both imbalanced and SMOTE-balanced conditions. The evaluation framework prioritised recall as the primary metric, grounded in the life-safety requirement that genuine tsunami-generating earthquakes must not be missed.

The key conclusions are:

- Logistic Regression achieved the highest recall of 0.800 without SMOTE and 0.780 with SMOTE, making it the recommended model for operational tsunami early warning.
- Random Forest demonstrated the strongest overall discriminative performance (ROC-AUC 0.913, F1 0.203) and showed the greatest recall improvement from SMOTE application (0.450 to 0.610).
- SMOTE consistently improved recall for three of the four models, with diminishing returns for Logistic Regression which already uses cost-sensitive class weighting.
- Threshold tuning on Logistic Regression can raise recall up to 0.97 at a threshold of 0.05, enabling near-comprehensive detection.
- The aggregated confusion matrix confirmed that the recommended model correctly identifies 97 out of 120 genuine tsunami events, with only 23 misses.

The proposed system requires only standard seismic parameters automatically computed within seconds of earthquake detection; it operates at negligible computational cost; and it is transparent enough to be audited and trusted by domain experts.

9.2 Future Scope

Expansion to additional tsunami triggers: Future iterations should incorporate submarine landslide and volcanic eruption events as additional trigger types.

Adaptive spatio-temporal windows: Future work should explore magnitude- and depth-dependent adaptive windows that more precisely reflect tsunami propagation physics.

Deep learning architectures: LSTM networks can exploit temporal patterns across sequences of seismic events, while CNN architectures can extract spatial features from gridded seismic intensity maps.

Real-time prediction pipeline: The system should be extended into a streaming pipeline subscribing to the live USGS Earthquake Notification Service (ENS) API, evaluating each incoming seismic event in real time.

Hybrid ML-physics integration: Machine learning performs rapid first-pass triage within seconds of detection, with physics-based numerical simulation triggered only for high-risk events.

Explainability integration: Incorporating SHAP or LIME would allow each prediction to be decomposed into feature-level contributions, increasing operator trust.

Global multi-region validation: Future work should evaluate whether region-specific models trained on Circum-Pacific Belt or Mediterranean data provide superior recall compared to the global model.

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